AMTI (NMTC) - 2015

RAMANUJAN CONTEST - INTER LEVEL

PART - A

1. When x, y, z are real, the minimum value of $2x^2 + 2y^2 + 5z^2 - 2xy - 4xz - 4yz - 4x - 2x + 15$ is
(A) 18 (B) 25 (C) 10 (D) 15

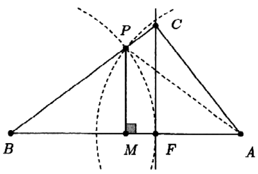
2. If f(x) is a real values function and (for any x real $\neq 0$) $3f(x) - 2f\left(\frac{f}{x}\right) = x$, the value of f(4) is
(A) $\frac{5}{2}$ (B) $\frac{7}{2}$ (C) $\frac{1}{2}$ (D) 0

3. The number of natural numbers n for which $(3n - 4)$, $(4n - 5)$, $(5n - 3)$ are all prime number is
(A) 0 (B) 1 (C) 2 (D) infinite

4. Five points on a circle are numbered 1, 2, 3, 4 and 5. A frog jumps in a clockwise direction from one point to another round the circle. If it is on an odd numbered point, it jumps one point, and if it is on an even–numbed point, it moves two points. The frog begins on point 5. After 2004 jumps it will be on point
(A) 1 (B) 2 (C) 3 (D) 4

5. The number of solutions of the equation $(\log_{10}x)^2 = \log_{10}(100x)$ is
(A) 0 (B) 1 (C) 2 (D) 4

6. ABC is a right angled triangle with hypotenuse AB and CF is the altitude. The circle with center at B and passing through F and another circle of the same radius with center at A insersect on the side BC. Then the ratio $\frac{BF}{BC}$ is equal to



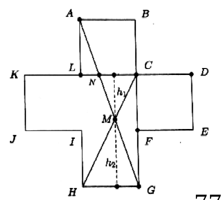
(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{\sqrt[3]{2}}$ (D) $\frac{1}{2\sqrt{2}}$

7. The number of natural numbers n, for which n! + 5 is a perfect cube is
(A) 0 (B) 1 (C) 2 (D) 5

8. Two sides of a triangle are 8 cm and 18 cm and the bisector of the angle formed by them is of length $\frac{60}{13}$ cm. The length of the third side (in cm) is
(A) 22 (B) 23 (C) 24 (D) 25

9. A number of workers in their tea time went to a tea shop and took tea and snacks. At the end they decided to split the bill evenly among them. If each contributes Rs. 16, they found that they were 4 Rs. short, while if each contributes Rs. 19, they had enough to pay the bill, 15% for the tip and Rs. 2 left over. The number of workers is
(A) 10 (B) 12 (C) 15 (D) 11

10. Consider the figure ABCDEFGHIJKL as shown. Each side is of length 4 and the angle between any two consecutive sides is a right angle. AG and CH cut at M. The area of the quadrilateral ABCM is



(A) $\frac{88}{5}$ (B) $\frac{44}{3}$ (C) $\frac{77}{9}$ (D) $\frac{62}{3}$

11. The number of pairs of natural numbers (x, y) which satisfy $\frac{5}{x}+\frac{6}{y}=1$ is

(A) 5 (B) 30 (C) 11 (D) 8

12. The sides of a triangle are 15, 20 and 25. The length of the shortest altitude is

(A) 6 (B) 12 (C) 10 (D) 13

13. a, b, c are reals such that a – 7b + 8c = 4 and 8a + 4b – c = 7. The value of $a^2 - b^2 + c^2$ is

(A) 0 (B) 12 (C) 8 (D) 1

14. n is a five digit number. If q and r are respectively the quotient and remainder when n is divided by 100, the number of n for which (q + r) is divisible by 11 is

(A) 8181 (B) 8180 (C) 8182 (D) 9000

15. A sphere is inscribed in a cube that has a surface area of 24 cm^2. A second cube is then inscribed within the sphere. The surface area of the inner cube in square centimeters is

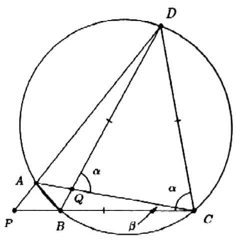
(A) 3 (B) 8 (C) 6 (D) 9

PART - B

16. The smallest multiple of 15 such that the result contains only 0 or 8 is ______.

17. Vishwa is walking up a stair that has 10 steps and with each stride the goes up either one step of two steps. The number of different ways Vishwa can go up the stars is ______.

18. The quadrilateral ABCD is inscribed in a circle. The diagonals AC and BD cut at Q. DA produced and CB produced cut at P. If CD = CP = DQ, then ∠DAC = ______.



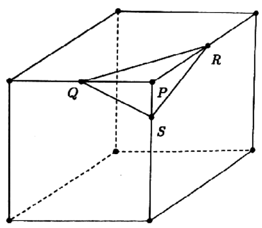
19. The sum of the first 100 terms of a arithmetic progression is –1, and sum of the 2^{nd}, 4^{th}, 6^{th}, 8^{th}, ... and the 100^{th} terms is 1. Then the sum of the squares of the first 100 terms of the A.P. is ______.

20. The number of pairs of positive integers (m, n) such that m, n have no factors greater that 1 and $\frac{m}{n}+\frac{14n}{9m}$ is an integer is ______.

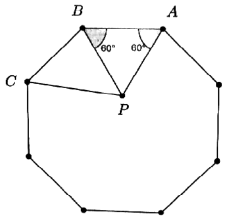
21. The number of two digit numbers that increase by 75% when their digits are reversed is _____.

22. ABCD is a rectangle. A is (14, –32), B is (2014, 168) and D is (10, y) for some integer y. The area of the rectangle is ______.

23. P is the vertex of cuboid. Q, R, S are points on the edges shown. If PQ = 4 cm, PR = 4 cm and PS = 2 cm and the area of triangle QRS is $\sqrt{K}$ cm^2 then K = ______.



24. ABCDEFGH is a regular octagon. ABP is an equilateral triangle with P inside the octagon. Then measure of 2∠APC = _______.



25. The number of numbers from 12 to 12345 inclusive having digits which are consecutive and in increasing order reading from left to right is _______.

26. The number of integer pairs (m, n) such that $15m^2 - 7n^2 = 9$ is _______.

27. A positive integer n is a multiple of 7. If $\sqrt{n}$ lies between 15 and 16, the number of possible values of n is ______.

28. The number 27000001 has exactly 4 prime factors. The sum of these prime factors is _____.

29. a is an integer such that $\frac{a}{23!} = 1 + \frac{1}{2} + \frac{1}{3} + ... + \frac{1}{23}$. The remainder when a is divided by 13 is ____.

30. ABC is a right triangle as shown. D is the midpoint of AC. E is a point on BC such that $\angle DEB = 30^0$. DE = K · AB where K is a number. Then the value of K is ______.